

ALLEGRETTI GIOVANNI

Senior C++ Software Engineer

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Rome (IT)

SUMMARY

Senior C++ Software Engineer with six years of experience in performance-critical development, Qt/QML desktop applications, and computer vision. Strong background in software architecture design and CI/CD integration. Passionate about robotics and embedded systems.

EXPERIENCE

DMA Accurate & Reliable - Software Engineer

Apr 2023 – Present, Turin (IT) - remote

- Lead architectural decisions on the real-time video inspection framework — execution model, threading strategy, and system composition — alongside API design and performance-critical core development.
- Designed the GPU-accelerated inference layer (TensorRT) and scaled it across multiple cards to meet throughput requirements of the most demanding deployments.
- Led the full redesign of the catenary wear measurement product: ground-up Qt/QML UI rewrite, generalization of the analysis from 1 to N wires, and rebuild of the multi-source data synchronization.
- Designed and built a standalone Qt/QML real-time monitoring application around the inspection framework — live camera streams, defect detection results, runtime benchmarks.
- Containerized the core libraries, packaged dependencies into a self-hosted Debian/APT repository, and built multi-platform CI/CD pipelines (Linux GCC + Windows MSVC).

ALTEN (consultant at Leonardo Aircraft) - Software Engineer

Jul 2020 – Mar 2023, Turin (IT)

- Designed and developed HMI applications in C++ and Qt for avionics mission simulation, enabling real-time monitoring of cockpit inputs and injection of simulated failures.
- Developed and maintained a flight and mission simulation model in C++, focusing on reliability, performance, and system integration.

ALTEC - Aerospace Logistics Technology Engineering Company - Software Engineer (Internship)

Mar 2019 – Jul 2019, Turin (IT)

- Designed, developed, and tested a C++ software module for planetary exploration robots, computing terrain-adaptive kinematics for a 6-DOF robotic arm. Integrated into an existing ROS/Gazebo framework, with roll/pitch error ≤ 0.10 rad and Z error ≤ 0.10 m on inclined slopes.

PROJECTS AND COLLABORATIONS

ROS Object Grasping — Cognitive Robotics

Università Degli Studi di Salerno

2019, Fisciano (IT)

- Designed and developed a cognitive robotics system for object recognition and manipulation using computer vision and neural networks.

Autonomous Vision Rover — Computer Vision

Università Degli Studi di Salerno

2019, Fisciano (IT)

- Developed an autonomous rover system for navigation in a controlled environment.

STM32 Differential-Drive Rover - Embedded Systems

Università Degli Studi di Salerno

2018, Fisciano (IT)

- Designed and implemented a custom PID closed-loop controller and firmware for both manual and autonomous (line-following) driving modes on an STM32F401RETx.

LANGUAGES

Italian

Native



English

Advanced



TECHNICAL SKILLS

Core

C++ Qt/QML Computer Vision

Multithreading Software Architecture

Design Patterns

Tools & CI/CD

Git GitLab CI/CD Docker CMake

GTest Valgrind Kestra

DebianPackaging vcpkg gdb Intel

VTune GitHub

Libraries & Frameworks

OpenCV Boost ZMQ ROS

TensorFlow TensorRT Gazebo

Platforms

Linux Windows Embedded (ESP32,

STM32) Nvidia Jetson

Other Languages

Python C MATLAB Bash

Methodologies

Agile/Scrum

PERSONAL TRAITS

Problem Solving Teamwork

Critical Thinking Attention to Detail

Continuous Learning

EDUCATION

Facial Expression Recognition - Computer Vision

Master thesis Erasmus Program - University of Groningen

2019, Groningen (NL)

- Automatic Facial Expression Recognition in unconstrained ('in-the-wild') video, using temporal modelling (CNN+LSTM, C3D) and multimodal video+audio fusion.

Master's Degree in Computer Engineering

110/110

Università Degli Studi di Salerno

2020, Fisciano (IT)

Bachelor's Degree in Computer Engineering

95/110

Università Degli Studi di Salerno

2017, Fisciano (IT)
